LabOS | A Transport Experiment That Calibrates and Learns

Device communication → autonomous calibration → live feedback → physics-guided next experiment

A / DEVICE COMMUNICATION

Commands down • evidence back

Central tools → bounded command

command_id / stage_id / device_id / deadline

PC-initiated HTTP(S)

Poll + heartbeat
Results + raw artifacts

PC i … PC N / local PC Agent

Device sessions + command journal + watchdog
Readbacks / flags / timestamps / local raw cache

Driver adapters / reviewed local code

VISA I/O or vendor SDK; SCPI only where supported

GPIB / USB / serial / TCP-IP

Lock-in master: sine OUT + Vxx(1f)

Lock-in follower: selected Vxy(2f)

attoDRY: T / B | 2400: Vg / Ig

Physical signal and reference cables

Sine OUT —Rs→ Sample + return

Master f_I → follower REF; select harmonic

Network timing and physical phase reference
are separate parts of the experiment.

B / MEASUREMENT AND FEEDBACK

1 Define the transport campaign

Sample / wiring / T, B, Vg / current and frequency envelope

2 Autonomous calibration

Verify actual current, reference and phase
Choose range / filtering; repeat baseline checks
Reuse valid calibration or issue a new calibration_id

3 Prepare + acquire the next point

Join actual T/B/Vg readiness; settle after setting changes
Recheck calibration validity at the observed state
Save Vxx(1f), Vxy(2f), X/Y, readbacks and timestamps

4 Live qualification

Reference lock / overload / freshness / drift / leakage
Recoverable QC issue → recalibrate; hard fault → pause

Qualified evidence

5 Central feedback + scientific analysis

Fit amplitude / phase / drive dependence; compare controls
Propose a repeat, a new setting or a discriminating test

6 Validate update / close the campaign

Limits + locks + versioned update / recorded closure

Objective met / budget reached → archive + close

C / SCIENTIFIC NEXT-STEP DESIGN

PLANNED • SERVER-LOCAL THEORY

Theory MCP / numerical jobs / versioned models

Receive a qualified-data snapshot

X/Y, I, f, T/B/Vg + calibration + snapshot IDs

Run models + estimate uncertainty

Transport / circuit / thermal-response hypotheses

Compare predictions with experiment

Residuals, validity range and competing explanations

Rank informative next measurements

Model discrimination / uncertainty / time / constraints

What changes after feedback?

Noisy but stable
Longer averaging / suitable filtering

Overload / reference drift
Correct range / reference; reacquire

Nonlinearity / possible heating
Lower drive; add frequency / return controls

Fault / uncertain command
Pause, local closure and state reconciliation

Every revision preserves raw evidence and passes through the same execution authority.

LOCAL DEVICE LOOPS

Reference / readout / local fault response
No LLM in the timing-critical path

CENTRAL POINT / CHUNK FEEDBACK

Event-driven QC and bounded stage revisions
Dashboard follows state and qualified data

PLANNED ASYNCHRONOUS THEORY

Compute budget + uncertainty + snapshot checks
A local LLM is not a numerical physics solver

Transport workflow design; the theory/compute coupling is planned. No measured curves or material mechanism claims are implied.